Comprehensive Summary: Campus Infrastructure, Buildings, and Resource Baseline

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CAMPUS INFRASTRUCTURE, BUILDINGS, AND RESOURCE BASELINE REFERENCE GUIDE

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Authoring Directorate: Campus Decarbonization Taskforce and Facilities Management (DEP_ADMN).

Subject: Comprehensive Master Reference for Institutional Decarbonization.

Institution: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai, Maharashtra, India.

1. INSTITUTIONAL METRICS AND GEOGRAPHIC LOCATION

Vivekananda Education Society's Institute of Technology (VESIT Green Campus) is an autonomous higher education engineering institution situated in the eastern suburb of Chembur, Mumbai, Maharashtra, India (Latitude: 19.0760° N, Longitude: 72.8777° E). Established on a contiguous 32,500 m2 built-up campus, the college serves 3,000 undergraduate and postgraduate engineering students, supported by 200 faculty and administrative staff members.

The campus baseline year is established as calendar year 2024, with a statutory commitment to achieve complete operational carbon neutrality (TRG_NET_ZERO) by December 31, 2040.

2. BUILDING INFRASTRUCTURE AND OCCUPANCY ASSETS

The institution comprises seven primary permanent buildings:

- Academic Block A (BLD_ACA): Built-up area 7,500 m2 | Occupancy: 1,100 | Primary use: Classrooms, tutorial halls, and faculty chambers | Housing: Department of Computer Engineering (DEP_CMPN, Head: Dr. N. Joshi) and Department of Electronics & Telecommunication (DEP_EXTC, Head: Dr. R. Kulkarni) | Solar PV installed: 80 kWp (200 modules) | 2024 Electricity: 382,500 kWh.

- Academic Block B (BLD_ACB): Area: 6,800 m2 | Occupancy: 950 | Primary use: Classrooms, seminar auditoria | Housing: Department of Information Technology (DEP_INFT, Head: Dr. S. Nair) | Solar PV: 70 kWp (175 modules) | 2024 Electricity: 333,200 kWh.

- Computer Laboratories (BLD_LAB): Area: 5,200 m2 | Occupancy: 650 | Primary use: 16 high-density computer labs, 2 primary server rooms, AI Research Center | Housing: Department of Artificial Intelligence & Data Science (DEP_AIDS, Head: Dr. P. Sharma) | Solar PV: None (roof obstructed by cooling chillers) | 2024 Electricity: 582,400 kWh (highest intensity: 112 kWh/m2/year).

- Library (BLD_LIB): Area: 3,100 m2 | Occupancy: 350 | Primary use: Reading rooms, digital archives, stacks | Solar PV: 30 kWp (75 modules) | 2024 Electricity: 168,200 kWh.

- Auditorium (BLD_AUD): Area: 2,800 m2 | Occupancy: 800 seats | Primary use: Conferences, keynotes, cultural events | 2024 Electricity: 134,200 kWh | Proposed Phase 2 Solar: 50 kWp bifacial array.

- Canteen (BLD_CAN): Area: 2,100 m2 | Occupancy: 400 | Primary use: Food prep, dining, 2,200 daily meals | 2024 Electricity: 192,000 kWh | Annual food waste: 28,728 kg | On-site 250 kg/day biogas plant.

- Administration Block (BLD_ADM): Area: 5,000 m2 | Occupancy: 200 | Primary use: Directorate, Council Chambers, Finance, Facilities (DEP_ADMN, Head: Mr. V. Deshmukh) | Solar PV: 40 kWp (100 modules) | 2024 Electricity: 228,000 kWh.

3. RESOURCE CONSUMPTION, TELEMETRY, AND EMISSION BASELINES

3.1 Grid Electricity and Renewable Generation:

- Total Campus Electricity Consumption: 2,362,400 kWh in 2024.

- Imported Grid Electricity: 2,050,000 kWh from Maharashtra State regional grid (Emission factor EF_ELEC_GRID = 0.82 kg CO2e/kWh).

- Scope 2 Emissions: 1,681,000 kg CO2e (68.61% of campus total).

- On-Site Solar Generation (INIT_SOLAR_PH1): 312,400 kWh generated across 220 kWp rooftop arrays (BLD_ACA, BLD_ACB, BLD_LIB, BLD_ADM), delivering a 13.22% renewable electricity penetration rate and abating 256,168 kg CO2e.

3.2 Stationary Fuel and Scope 1 Emissions:

- Diesel Generator Consumption: 18,400 litres consumed during grid interruptions (Emission factor EF_DIESEL_GEN = 2.68 kg CO2e/L).

- Scope 1 Emissions: 49,312 kg CO2e (2.01% of campus total).

3.3 Water Resources and Rainwater Harvesting:

- Total Municipal Piped Water Intake: 48,000,000 litres (131,500 litres/working day).

- Rainwater Harvesting (INIT_RWH_SYSTEM): 12,400,000 litres harvested from 18,000 m2 terrace catchments and stored in 450,000 L cisterns, replacing 25.83% of municipal water needs. An additional 26,800,000 litres recharged into the local aquifer via four injection borewells.

3.4 Solid Waste and Circular Diversion:

- Total Solid Waste Generated: 68,400 kg.

- Canteen Biogas & Composting (INIT_BIOGAS_CANTEEN): 19,200 kg of food waste processed into cooking gas, abating 22,464 kg CO2e.

- Recyclables Recovered: 19,800 kg paper and 1,850 kg certified e-waste recycled with Greenscape Solutions.

- Diversion Rate: 48.2% achieved against target TRG_ZERO_WASTE (85.0% by 2028).

3.5 Transportation Mobility:

- Transportation Footprint: 580,000 kg CO2e across 3,200 commuters.

- Modal Split: Suburban Rail 48%, BEST Bus 22%, Two-Wheelers 16%, Private Cars 6%, Walking/Biking 8%.

- Campus Electric Shuttle (INIT_ELECTRIC_SHUTTLE): Two 14-seater EV vans delivered 38,400 passenger-trips, abating 36,000 kg CO2e.

3.6 Gross Baseline Emissions:

Total Calendar Year 2024 Footprint: 2,450,000 kg CO2e (2,450 metric tonnes CO2e). Target TRG_NET_ZERO mandates linear reduction of 153,125 kg CO2e annually to reach net-zero by 2040.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_SMR_022**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*